HOUSAM KAK

Front Office Investment Technology | Senior AI & Software Engineer | Systems and Processes Architect | Mathematician | Dubai, UAE

I build the compilers for things that are not code.

housamkak.com | blog.housamkak.com | blockforge.housamkak.com | github.com/HousamKak
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PROFILE

Senior AI and software engineer in front office investment technology, and a systems and processes architect. Leads product and senior software delivery for an enterprise conversational AI pilot at Davidson Kempner Capital Management, an elite $37B+ New York multi-strategy hedge fund, after client delivery across requirements, data integration, reporting and release engineering at OmniVista Solutions. Independently designs and builds systems that turn a description into a working thing: an AI agent workspace, a programmable 3D world engine, offline-first retail software and AI developer tools. Mechanical engineer with a mathematics major by training.

EXPERIENCE

Davidson Kempner Capital Management 2024 to present

Senior AI & Software Engineer, Front Office Investment Technology

Elite $37B+ NYC multi-strategy hedge fund | Product lead, enterprise conversational AI pilot | Client assignment through OmniVista Solutions

- Head the pilot AI systems team, owning product direction and senior software delivery for a conversational platform that answers staff questions from company data.
- The platform is used by 200+ internal users to find company information and support operational workflows.
- Connect enterprise data through MCP integrations, Gremlin-based graph access over Azure Cosmos DB and Azure AI Search, running on distributed microservices.
- Prioritise which data sources to connect and keep access boundaries intact as integrations grow.

OmniVista Solutions Apr 2023 to present

Consultant (joined as Associate Consultant)

Financial technology consultancy, Dubai | Client engagements for investment firms

- **OrbiMed:** led requirements and authored the BRD/TRD for an analyst recommendation portal; designed its architecture and trade submission to the Charles River order management system, and coordinated planning and CI/CD. Contributed to a reported 65% shorter trade-idea turnaround.
- **Davidson Kempner, data integration:** built C#/ASP.NET Core Windows services for ESG risk and market-data retrieval, released through Bitbucket, TeamCity and Octopus Deploy; traced data and optimised queries, cutting portfolio application load time by about 70%; designed Power BI reporting for Expected and Target Exposure.
- **Moore Capital Management:** led client support operations across 2,500+ tickets, automating support-data retrieval and analysis with Python and SQL.
- **HighTower Advisors:** automated Excel/VBA dashboards and SQL extraction and import preparation for trade, security, position and fund data.

Atlagene 2026 to present

Co-Founder and CTO (Engineering)

Consumer and B2B genomics startup | Co-founded with Yasser Shkeir (Co-Founder and CEO), who designed and built the platform end to end

- Cover engineering and infrastructure, accelerator programs and partner execution in Lebanon; wrote the B2B-first strategy and pitch, and built an interactive Three.js exhibition booth that paints in about 2 s instead of about 10 s on emulated 4G.

SELECTED SYSTEMS

Independent work. Case studies at housamkak.com/work.

H | AI agent workspace housamkak.com/work/h-system

TypeScript, React, SQLite, Tauri, MCP, Claude Agent SDK

A canvas where AI coding agents, terminals, generated applets and processes are tiles joined by typed wires on an event-sourced TypeScript backend; agents discover what they are wired to and drive it. Sole designer and engineer, in daily use as a supervised alpha.

BlockForge | Programmable voxel world engine housamkak.com/work/blockforge | blockforge.housamkak.com

C++17, SDL2, JavaScript, Three.js, Luau/WASM, MCP

A world compiler: scripts, generators and Luau mods compile into a voxel world that animates and films reproducibly. Browser and native C++17 clients on one format, versioned worlds with diff and merge, CLI and MCP control for agents. Sole designer and engineer.

Store POS ecosystem | Offline-first retail software housamkak.com/work/pos-store-management

Node.js, Express, SQLite, Turso, Cloudflare R2, PostgreSQL

Keeps a Lebanese store selling offline: local SQLite as source of truth, transactional checkout, FIFO lots, USD/LBP books, same-transaction audit records and outbox replication to Turso. Sole developer.

AnyThyng | A sentence becomes a working phone app housamkak.com/work/anythyng | anythyng.app

TypeScript, Claude Agent SDK, Kotlin/Compose, AWS, OpenTofu

Describe something and the phone screen becomes it: a Claude Agent SDK server with a default-deny tool boundary, sandboxed agent-written endpoints, a sealed Kotlin/Compose WebView shell and AWS infrastructure in OpenTofu with OIDC pipelines. Sole engineer; beta.

UIpen | Point-and-fix UI tool for coding agents housamkak.com/work/uipen | uipen.dev

TypeScript, React/Next.js, Bun, MCP

A dev-mode overlay: each pin on a running web app reaches Claude Code with its selector, source file, line and component, so the agent edits the right code. Seven framework source walkers, Ed25519 licensing, version 0.3.1. Sole designer and engineer.

Also: Studio (script-to-film pipeline), App Monitor (self-hosted crash reporting), Budgetability (multi-currency finance, live), Partforge and Electrical Lab (agent toolchains for parts and circuits), and Flying Screen (physics co-design engine and paper). Full index at housamkak.com/work.

SKILLS

AI and architecture: Conversational AI, Model Context Protocol (MCP), Claude Agent SDK, agent orchestration, Gremlin, Azure Cosmos DB, Azure AI Search, distributed microservices
Engineering: TypeScript, JavaScript, Python, C#, C++, SQL, Kotlin; Node.js, React, Next.js, ASP.NET Core, Tauri, Three.js
Data: PostgreSQL, SQL Server, SQLite, Turso, MySQL, Supabase
Cloud and DevOps: Azure, AWS, Cloudflare, OpenTofu, Docker, GitHub Actions, TeamCity, Octopus Deploy, Git
Product and consulting: Requirements elicitation, BRD/TRD, stakeholder management, technical planning, Power BI, Excel/VBA

EDUCATION

American University of Beirut 2018 to 2022

B.E. Mechanical Engineering, major in Mathematics (AUB)

Graduation project: autonomous miniature racing car with camera-based lane perception and pure-pursuit trajectory control. Mastercard Foundation Scholar (full undergraduate scholarship).

SE Factory, Beirut Aug to Dec 2022

Full-stack software engineering program (14-week intensive)

RECOGNITION AND LANGUAGES

Awards: 2021 Resolution Social Venture Challenge: co-founded Zari3tech, an agricultural social venture, and won a Resolution Project fellowship with $5,000 in funding; AUB SUMO Robotics Competition: second place, with a LEGO robot programmed in LabVIEW.
Languages: Arabic (native), English (full professional proficiency)